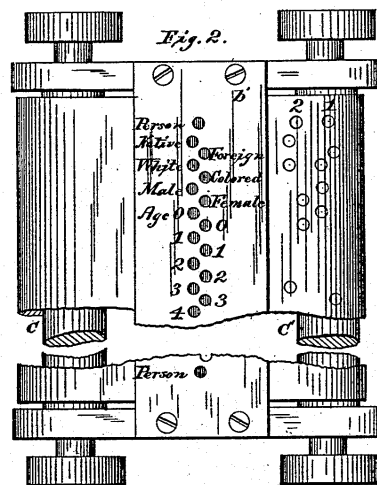
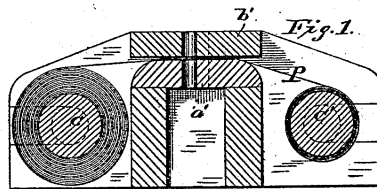




Please write your name in the box below.

Basic Algebra I Final Exam

12 March 2025, 12.30 PM-2.30 PM

<i>Weight</i>	<i>Problem</i>
5	1(i)
5	1(ii)
5	1(iii)
10	1(iv)
5	2(i)
5	2(ii)
5	2(iii)
5	2(iv)
5	3(i)
10	3(ii)
10	3(iii)
5	3(iv)
10	4
5	5(i)
10	5(ii)
100	

Problem 1. Let G be a group. For every element $g \in G$, one considers the function

$$\text{mult}_g : G \rightarrow G \quad x \mapsto g \cdot x.$$

- (i) Let $e \in G$ denote the identity element. What is mult_e ?

[No justification needed—do not write more than a few words.]

- (ii) Show for all $g, g' \in G$ that

$$\text{mult}_g \circ \text{mult}_{g'} = \text{mult}_{g \cdot g'}.$$

- (iii) Show for all $g \in G$ that the function mult_g is bijective.

(iv) Show that the function

$$\mu : G \rightarrow \text{Sym}(G) \quad g \mapsto \text{mult}_g$$

is an injective group homomorphism.

Problem 2. Let $H \subset G$ be a subgroup of a group.

(i) In class we defined the set G/H to be the quotient $G/\sim_H$ of G by a certain equivalence relation $\sim_H$ on G . Give the definition of $\sim_H$.

(ii) Define the *left coset* gH of an element $g \in G$. State the relation between gH and the equivalence class of g with respect to $\sim_H$.

(iii) Let $g \in G$ be an element. Construct a bijection between gH and H .

(iv) Suppose now that G is finite. Deduce that

$$|G| = |H| \cdot |G/H|.$$

Problem 3. Let $N \subset G$ be a subgroup of a group.

(i) What does it mean to say that N is a *normal subgroup*?

(ii) Suppose for the remainder of this problem that N is a normal subgroup. Show that there is a binary operation

$$G/N \times G/N \rightarrow G/N \quad (x, y) \mapsto x \cdot y$$

with the property that

$$[g] \cdot [g'] = [g \cdot g'] \quad \text{for all } g, g' \in G.$$

- (iii) Show that this binary operation makes G/N into a group, and that the quotient map $\pi : G \rightarrow G/N$ is a group homomorphism.

- (iv) Let $N \subset G$ be a normal subgroup, and let $\phi : G \rightarrow G'$ be a group homomorphism. State the criterion for there to exist a homomorphism

$$\psi : G/N \rightarrow G'$$

satisfying $\psi \circ \pi = \phi$.

Problem 4. Let G and G' be finite groups. Suppose that the order of G and the order of G' are coprime. Show that every homomorphism $\phi : G \rightarrow G'$ is trivial.

Problem 5. This problem is about the $(7,4)$ Hamming code constructed from the generator polynomial

$$g(X) = X^3 + X^2 + 1.$$

- (i) Compile the Zech logarithm table for $\mathbf{F}_2[X]/g\mathbf{F}_2[X]$ with base X .

(ii) Determine whether or not the message

$$X^5 + X^4 + X^3 + X^2 + X + 1$$

contains any errors. If so, correct the error and decode the message.